

RJ45 Cable Tester

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Described here is a simple RJ45 cable tester circuit which can be used for testing the RJ45 network cables. The circuit can check both straight-through and crossover-type RJ45 network cables. This is a low-cost tester designed using easily available components. Fig. 1 shows the difference between straight-through and crossover network cables. The tester also indicates the type of cable

under test with different sequences in which LEDs glow for both types of cables.

Circuit and working

Fig. 2 shows the circuit diagram of RJ45 cable tester. The circuit makes use of easily available components such as timer NE555 (IC1), decade counter CD4017B (IC2) and a few other components.

IC1 is a popular timer, wired in astable multivibrator mode generating output pulses of around 1Hz at pin 3.

These pulses serve as a clock for decade counter IC2. The outputs O0 through O3 of IC2 go high one after another with the input pulses at pin 14. These are

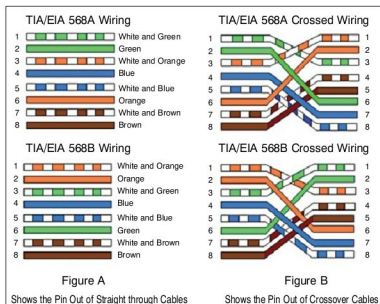


Fig. 1: Straight-through and crossover network cables

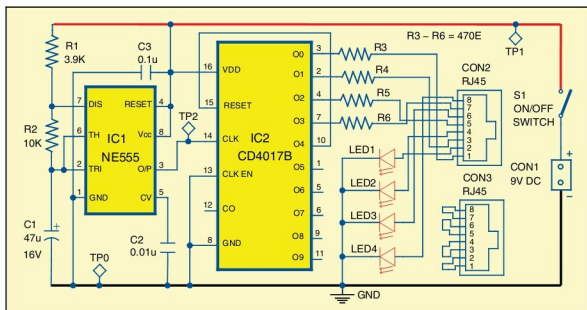


Fig. 2: Circuit diagram of RJ45 cable tester

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- IC2 - CD4017B decade counter
- LED1-LED4 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 3.9-kilo-ohm
- R2 - 10-kilo-ohm
- R3-R6 - 470-ohm

Capacitors:

- C1 - 47 μ F, 16V electrolytic
- C2 - 0.01 μ F ceramic disk
- C3 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal block connector
- CON2, CON3 - RJ45 connector
- S1 - On/off switch
- 16-pin IC base
- 8-pin IC base

TABLE I Sequence for Straight-through and Crossover Cables

Straight-through	Crossover
LED1	LED2
LED2	LED3
LED3	LED1
LED4	LED4

TABLE II Test Points

Test point	Details
TP0	GND
TP1	9V (when S1 on)
TP2	Around 1Hz

connected to the RJ45 socket CON2 such that a pulse is sent through one leg of each wire-pair and the return legs are connected to ground via LEDs showing the current flow in each pair. Resistors R3 through R6 are used to limit the current through each LED. Output O4 resets the counter IC2 on every fifth clock.

For testing the RJ45 cable, connect the cable between CON2 and